

Suraj Rao

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EDUCATION

University of Maryland College Park, United States

Sep 2025 - May 2027

Master of Science in Applied Machine Learning [GPA: 4/4]

Courses: Machine Learning, Deep Learning, AI Algorithms, NLP, Computer Vision, Statistics, Data Science and Analytics

Indian Institute of Information Technology Pune, India

Aug 2020 - Aug 2024

Bachelor of Technology in Computer Science and Engineering [GPA: 8.09/10]

Courses: Object Oriented Programming, Data Structures & Algorithms, Databases, Image Processing, Operating Systems

EXPERIENCE

AI Research Associate Intern | Mantra Softech, Bangalore, India

Dec 2023 – Jun 2024

- Optimized video surveillance pipeline by deploying real-time YOLOv8 for people detection and counting
- Reduced model size 4× via quantization, enabling parallel deployment across 4 camera streams using multiprocessing
- Increased throughput from 10 to 35 FPS via frame filtering, batching, and model profiling; deployed on RSC-101 [\[Product\]](#)
- Evaluated DALL·E Mini, Stable Diffusion, CLIP to build a text-to-video retrieval system for ~300+ hours of video footage

AI Research Associate | IIT Delhi, Delhi, India

Oct 2024 – Apr 2025

- Contributed to occluded vehicle detection by evaluating transformer-based vision models in Indian road scenarios
- Curated high-occlusion (>50%) data from RSUD20K, IDD, BDD100K via pseudo-labeling and built a Tkinter GUI for false positive correction, reducing annotation time from 7 min to 15 sec (~96%) [\[GitHub\]](#)
- Fine-tuned 4 SOTA transformer detectors (DETR, DINO, Grounding DINO, Co-DETR) on 4 NVIDIA GPUs using distributed multi-GPU training and benchmarked them using mAP and recall.

AI Research Assistant | UMD MIRAGE Lab, Maryland, United States

Jan 2026 – Present

- Built end-to-end lattice defect detection pipeline using YOLO26 on ~21K images across 3 materials for edge deployment
- Addressed 1:9 class imbalance using data augmentation and PyTorch weighted sampling, improving minority class detection
- Improved thickness classification by analyzing failure cases and refining training strategies, reducing false positives by 45%
- Achieved 96.3% mAP across 6 classes and deployed on NVIDIA Jetson using ONNX and TensorRT for real-time inspection

PROJECTS

Multimodal Semantic Search Engine | PyTorch, CLIP, FAISS, CUDA, Streamlit, HuggingFace

[\[GitHub\]](#)

- Built a multimodal semantic search engine using CLIP for text-to-image and image-to-image retrieval on 5.4K+ images
- Generated CLIP embeddings and indexed them with FAISS, improving search speed by 40% and enabling <200 ms latency
- Developed a Streamlit app with text and image query support, returning top-5 semantically relevant results
- Enhanced usability using HTML/CSS and deployed a scalable, containerized pipeline on AWS SageMaker

Smart Crop Yield Prediction | XGBoost, RandomForest, SVM, Flask

[\[GitHub\]](#)

- Built a crop yield prediction system for 60+ crops using multi-year agricultural data (2007–2020) across India
- Trained and benchmarked Random Forest, Support Vector Regression, XGBoost using R^2 and MAE with cross-validation
- Achieved peak R^2 of 0.965 and reduced prediction error by 25% (MAE: 0.895 → 0.667) by selecting XGBoost
- Deployed the model using a Flask web application for real-time yield prediction with ~700 ms latency

Low-Latency Face Detection using Haar Cascade | OpenCV, Haar Cascade, Fast R-CNN, YOLO

- Built a real-time face detection system using Haar Cascade classifier with OpenCV for webcam-based inference
- Analyzed Fast R-CNN, MobileNet-SSD, and YOLO to implement a lightweight, low-latency detection model
- Achieved real-time detection at ~30 FPS with consistent face localization under varying lighting conditions

[\[GitHub\]](#)

SKILLS

Programming: Python, C/C++, SQL, HTML, CSS

Libraries: PyTorch, NumPy, Pandas, Scikit-Learn, Matplotlib, OpenCV, Tkinter, Flask

Systems: CUDA, ONNX, TensorRT, OpenVINO, Parallel Programming, Distributed Training, Edge AI

AI & ML: CNN, Transformers, LLM, RAG, Deep Learning, Computer Vision, CLIP, HuggingFace

Tools: Git, Linux, Docker, Streamlit, FAISS, Azure, AWS SageMaker, MLflow, CI/CD, TensorBoard